

Week 3 Workshop

The excel file ff_25.xlsx contains 25 columns (portfolios) and the excel file rf.xlsx contains the risk-free rate. The sample period is monthly Jan 1963 to Dec 2013.

The 25 Fama–French portfolios are formed by sorting stocks in two ways. First, firms are sorted by size (market capitalisation) into five groups, shown by the rows from Small to Big. Second, firms are sorted by book-to-market ratio (B/M) into five groups, shown by the columns from Low to High. Each cell in the 5×5 table therefore represents a portfolio of firms with a particular combination of size and book-to-market characteristics. In the following table, col x refers to the x column in the excel file.

		BtM				
		Low	2	3	4	high
size	small	col 1	col 2	col 3	col 4	col 5
	2	col 6	col 7	col 8	col 9	col 10
	3	col 11	col 12	col 13	col 14	col 15
	4	col 16	col 17	col 18	col 19	col 20
	Big	col 21	col 22	col 23	col 24	col 25

The following table reports the sample mean excess return (monthly), Jan 1963 to Dec 2013. This is the classic Fama table.

		BtM				
		Low	2	3	4	high
size	small	0.30	0.81	0.85	1.03	1.17
	2	0.48	0.73	0.93	0.96	1.04
	3	0.50	0.79	0.79	0.89	1.09
	4	0.60	0.57	0.72	0.87	0.87
	Big	0.46	0.52	0.49	0.57	0.65

How to read the table?

- a. Each number is an average monthly return (%).
- b. Rows go from small firms at the top to big firms at the bottom.
- c. Columns go from low B/M (growth) on the left to high B/M (value) on the right.
- d. Note: Low B/M = growth stocks and High B/M = value stocks
- e. So, for example:
 - (i) 0.30 in the top-left means small growth stocks earned an average of 0.30% per month
 - (ii) 1.17 in the top-right means small value stocks earned an average of 1.17% per month

Main patterns

1. There is a strong value effect

For almost every size group, mean return tends to increase as B/M increases from left to right. For example,

- Small firms: $0.30 \rightarrow 0.81 \rightarrow 0.85 \rightarrow 1.03 \rightarrow 1.17$
- Big firms: $0.46 \rightarrow 0.52 \rightarrow 0.49 \rightarrow 0.57 \rightarrow 0.65$

This suggests that value portfolios tend to earn higher average returns than growth portfolios.

2. There is also a size effect

Looking down many columns, returns tend to be higher for smaller firms than for bigger firms, especially in the middle and high B/M columns.

For example:

- High B/M column: 1.17, 1.04, 1.09, 0.87, 0.65
- Next-highest B/M column: 1.03, 0.96, 0.89, 0.87, 0.57

This suggests that small stocks often outperform big stocks, especially among value stocks.

3. The highest returns are in small-value portfolios

The largest number in the table is 1.17, which is the small, high B/M portfolio. **This is the classic Fama–French result: small + value tends to have the highest average return.**

4. The lowest returns are in growth portfolios, especially small growth

The smallest number is 0.30, for small, low B/M stocks.

So the weakest-performing group here is small growth.

Overall interpretation

This table is consistent with the usual Fama–French size and value premiums:

- Value stocks outperform growth stocks
- Small stocks often outperform big stocks
- Small value stocks have the highest average returns